Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 32. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the f1 score by dataset.

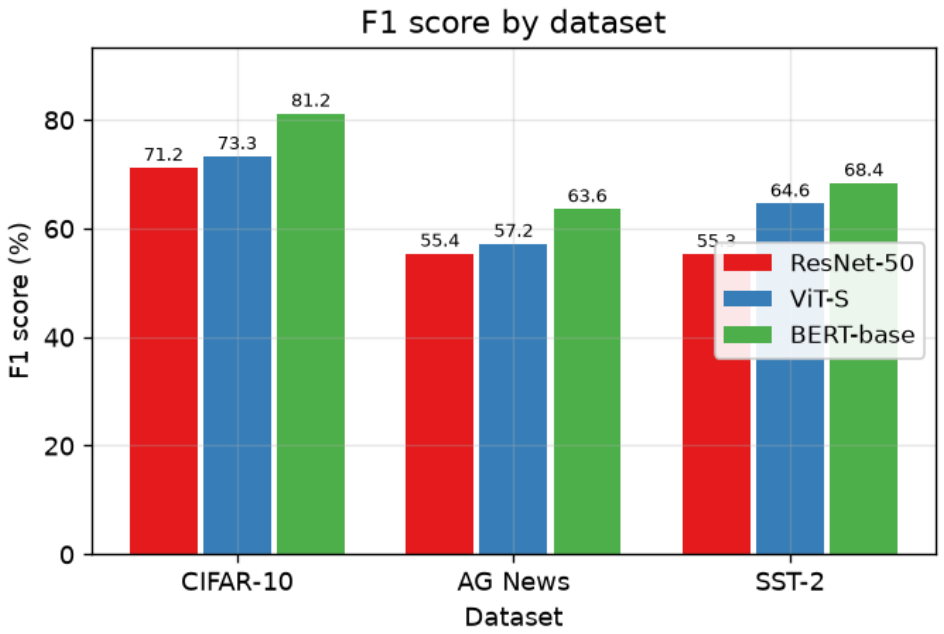


Figure 1: F1 score by dataset for the models in this experiment.

As shown in Figure 1, on AG News, ViT-S obtains a F1 score of 57.2%. On SST-2, BERT-base obtains a F1 score of 66.5%. As shown in Figure 1, on SST-2, ResNet-50 obtains a F1 score of 64.6%. On AG News, BERT-base obtains a F1 score of 110.3% (Figure 1).

Figure 2 shows the training loss.

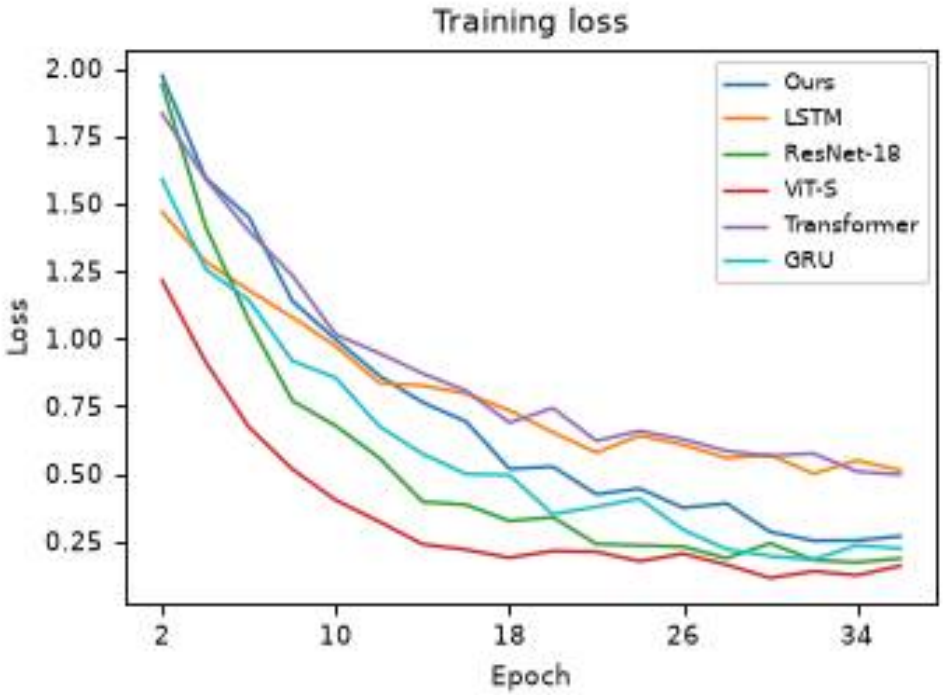


Figure 2: Training loss for the models in this experiment.

LSTM has a loss of 0.512 at epoch 36. As shown in Figure 2, at epoch 28, the loss of ViT-S is over 0.1. At epoch 26, the loss of ViT-S is -0.206.

Figure 3 shows the validation accuracy during training.

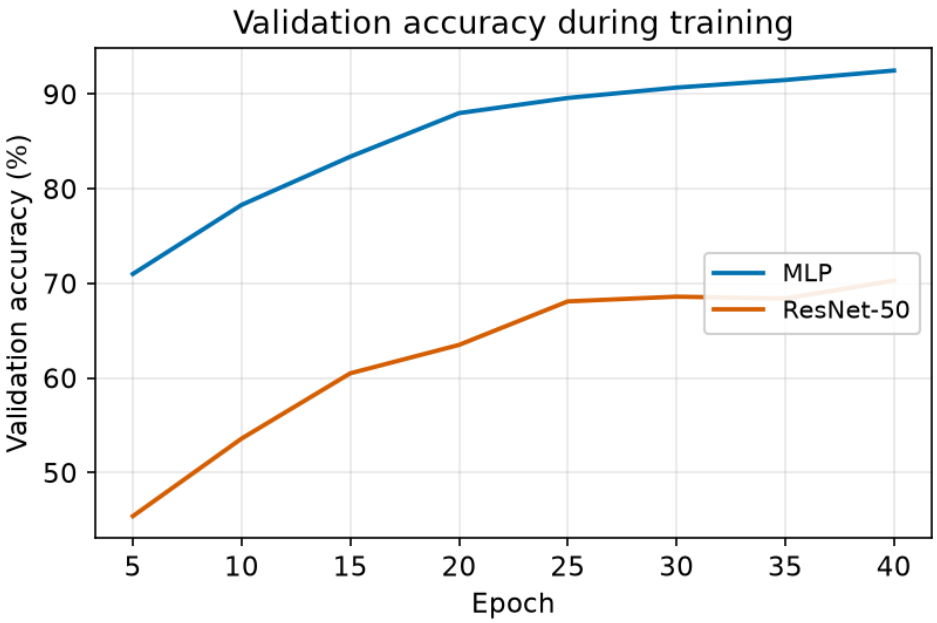


Figure 3: Validation accuracy during training for the models in this experiment.

MLP attains a validation accuracy of 91.5% at epoch 35. As shown in Figure 3, ResNet-50 attains a validation accuracy of about 60% at epoch 20. At epoch 30, MLP achieves 101.1% validation accuracy. ResNet-50 reaches 68.1% validation accuracy at epoch 25 (Figure 3).

Figure 4 shows the validation accuracy by dataset.

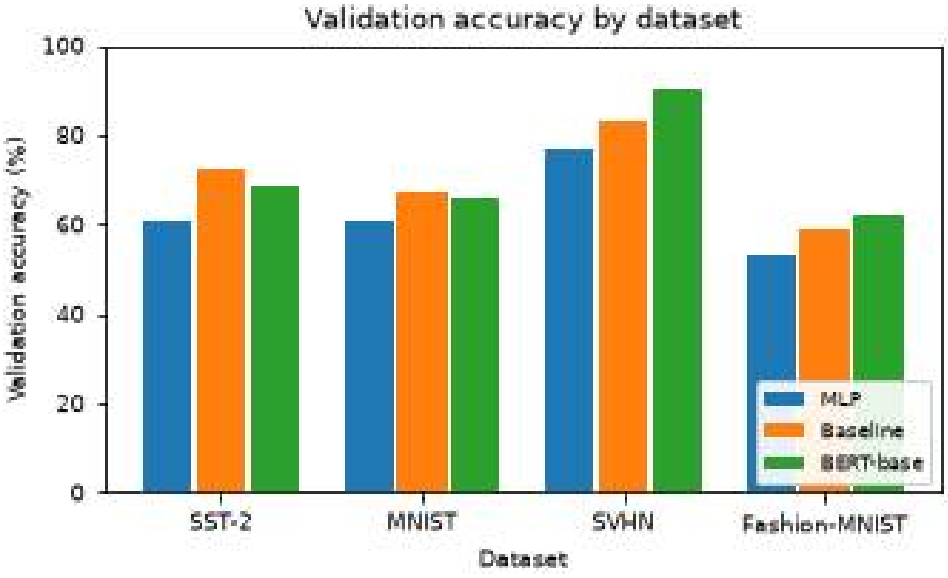


Figure 4: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 4, on Fashion-MNIST, BERT-base is 3 points better than Baseline. As shown in Figure 4, MLP reaches about 60% validation accuracy on MNIST. On SST-2, BERT-base is 10.1 points better than MLP (Figure 4). On MNIST, MLP obtains a validation accuracy of about 60%.

3 Discussion

Logistic Regression reaches 79.2% accuracy on IMDB. Overall, the results are in line with what we expected from the literature.